Year 10 Mathematics

Number 31 Positive and Negative Indices

6.1 Positive and Negative Indices1 Positive and Negative Indices

Positive and Negative Indices

STARTER · 5 MIN

Date: _____ **Title:** Positive and Negative Indices

LEARNING OBJECTIVES

- Write and work out powers using index notation
- Use the laws of indices to multiply and divide powers
- Understand and work out negative indices

STARTER *Answer in your book. No calculators.*

1. Work out 4×4
2. Work out $2 \times 2 \times 2$
3. Work out $1 \div 5$
4. Write $1 \div 100$ as a fraction

Starter — Answers

STARTER

Tick or correct in green pen

1. $4 \times 4 = \mathbf{16}$

2. $2 \times 2 \times 2 = \mathbf{8}$

3. $1 \div 5 = \mathbf{\frac{1}{5}}$

4. $1 \div 100 = \mathbf{\frac{1}{100}}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can write and work out powers using index notation	☐
SILVER	I can use the laws of indices to multiply and divide powers	☐
GOLD	I can understand and work out negative indices	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Indices

INTRODUCTION · 3 MIN

The index (power) tells you how many times the number is multiplied by itself.

Index notation

Base and index.

$2^4 = 2 \times 2 \times 2 \times 2 = 16$

Multiply and divide

Same base: add or subtract the indices.

$3^5 \times 3^2 = 3^7, \quad 3^5 \div 3^2 = 3^3$

Zero index

Anything to the power 0 is 1.

$7^0 = 1$

Negative index

A negative index means 'one over'.

$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$

Examples 1 and 2 — Positive Indices

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Work out

3^4

Example 2

Write using index notation:

$5 \times 5 \times 5 \times 5 \times 5 \times 5$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$3^4 = 3 \times 3 \times 3 \times 3$

$= 81$

Tip: the index tells you how many 3s to multiply.

Example 2

Count the 5s: there are 6

$= 5^6$

Tip: the base is the number being multiplied.

Task 1

PRACTICE · 9 MIN

BRONZE

Answer in your book

1 Work out 2^5

2 Work out 4^3

3 Work out 10^4

4 Write in index form: $7 \times 7 \times 7 \times 7$

5 Work out 6^0

★ Work out $2^3 \times 3^2$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $2 \times 2 \times 2 \times 2 \times 2 = \mathbf{32}$

2. $4 \times 4 \times 4 = \mathbf{64}$

3. $10 \times 10 \times 10 \times 10 = \mathbf{10\,000}$

4. $7 \times 7 \times 7 \times 7 = \mathbf{7^4}$

5. $6^0 = \mathbf{1}$

★ $8 \times 9 = \mathbf{72}$

Examples 3 and 4 — Multiplying and Dividing Powers

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

Example 3

Simplify, leaving your answer as a power:

$4^3 \times 4^5$

Example 4

Simplify, leaving your answer as a power:

$6^9 \div 6^2$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$4^3 \times 4^5 = 4^{3+5}$

$= 4^8$

Tip: multiply: add the indices.

Example 4

$6^9 \div 6^2 = 6^{9-2}$

$= 6^7$

Tip: divide: subtract the indices.

Task 2

PRACTICE · 9 MIN

SILVER

Simplify, leaving each answer as a power

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

1. $2^4 \times 2^3$

2. $5^8 \div 5^3$

3. $9^2 \times 9^6$

4. $3^{10} \div 3^4$

5. $7^5 \times 7$

6. $(2^3)^2$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $2^{4+3} = \mathbf{2^{7}}$

2. $5^{8-3} = \mathbf{5^{5}}$

3. $9^{2+6} = \mathbf{9^{8}}$

4. $3^{10-4} = \mathbf{3^{6}}$

5. $7^{5+1} = \mathbf{7^{6}}$

6. $2^{3\times 2} = \mathbf{2^{6}}$

Examples 5 and 6 — Negative Indices

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $a^{-n} = \frac{1}{a^n}$

Example 5

(1 mark)

Work out

4^{-2}

Example 6

(2 marks)

Simplify, then work out:

$2^3 \div 2^5$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$4^{-2} = \frac{1}{4^2}$$

$$= \frac{1}{16}$$

Tip: negative index: one over.

Example 6

$$2^3 \div 2^5 = 2^{3-5}$$

$$= 2^{-2}$$

$$= \frac{1}{4}$$

Tip: the index can be negative after subtracting.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $a^{-n} = \frac{1}{a^n}$

1 Work out 3^{-2} (1 mark)

2 Work out 10^{-3} (1 mark)

3 Work out 5^{-1} (1 mark)

4 Simplify, then work out $3^4 \div 3^6$ (2 marks)

5 Write as a power of 2: $\frac{1}{8}$ (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $\frac{1}{3^2} = \mathbf{\frac{1}{9}}$

2 $\frac{1}{10^3} = \mathbf{\frac{1}{1000}}$

3 $5^{-1} = \mathbf{\frac{1}{5}}$

4 $3^{4-6} = 3^{-2} = \mathbf{\frac{1}{9}}$

5 $\frac{1}{2^3} = \mathbf{2^{-3}}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Work out 2^6

SILVER Simplify $8^5 \div 8^2$

GOLD Work out 2^{-4}

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $2 \times 2 \times 2 \times 2 \times 2 \times 2 = \mathbf{64}$

SILVER $8^{5-2} = \mathbf{8^3}$

GOLD $\frac{1}{2^4} = \frac{\mathbf{1}}{\mathbf{16}}$

Exam Practice — Index Notation

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

(a) Write $2 \times 2 \times 2 \times 2 \times 2$ in index form. (1)

.....

(b) Work out the value of (1)

3^4

.....

Question 2

(3 marks)

(a) Write $7 \times 7 \times 7 \times a \times a$ in index form. (1)

.....

(b) Work out the value of (2)

$2^3 \times 5^2$

.....

Exam Practice — Laws of Indices

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

Question 3

(2 marks)

Simplify. Leave your answers in index form.

(a) (1)

$6^3 \times 6^5$

.....

(b) (1)

$10^8 \div 10^3$

.....

Question 4

(3 marks)

(a) Find the value of n. (1)

$2^7 \times 2^n = 2^{12}$

n = ...

(b) Work out the value of (2)

$2^9 \div 2^6$

.....

Exam Practice — Negative Indices

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $a^{-n} = \frac{1}{a^n}$

Question 5

(2 marks)

Give your answers as fractions.

(a) Write down the value of (1)

5^{-1}

.....

(b) Work out the value of (1)

2^{-3}

.....

Question 6

(3 marks)

(a) Write as a single power of 3 (1)

$3^4 \div 3^7$

.....

(b) Hence write down the value of (2)

$3^4 \div 3^7$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 2^5 — B1

(b) 81 — B1

Tip: $3^4 = 3 \times 3 \times 3 \times 3$, not 3×4.

Question 2

(a) 7^3a^2 — B1

(b) 8×25 — M1

= 200 — A1

Tip: work out each power first.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 6^8 — B1

(b) 10^5 — B1

Tip: same base → add or subtract the powers.

Question 4

(a) n = 5 — B1

(b) 2^3 — M1

= 8 — A1

Tip: simplify first, then evaluate.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $\frac{1}{5}$ — B1

(b) $\frac{1}{8}$ — B1

Tip: a negative power means 'one over'.

Question 6

(a) 3^{-3} — B1

(b) $\frac{1}{3^3}$ — M1

$= \frac{1}{27}$ — A1

Tip: $a^{-n} = 1 \div a^n$.